

Banking System API Documentation

Overview

Welcome to the Banking System API documentation. This API serves as the backbone of the Bank System platform, enabling seamless account and transaction management. It provides a range of endpoints for creating and managing user accounts, and tracking financial activities. Built with scalability and security in mind, the API supports robust error handling, logging, and integration with modern banking workflows.

Designed for developers, it includes detailed request and response structures, offering flexibility for integration into diverse applications. Whether for personal finance tools, corporate banking solutions, or payment platforms, this API facilitates efficient and reliable financial operations.

API Information

· **Title:** Banking System API

· **Description:** This API serves as the backbone of the Bank System platform, enabling seamless account and transaction management.

· **Contact:** Ahmed Bassem Ramzy (rameya683@gmail.com)

· **Version:** v1

Endpoints

User Management

Get User by ID

- **Endpoint:** /api/users/{userId}
- **Method:** GET
- **Summary:** Gets a user by their ID.
- **Parameters:**
 - userId (path, required): The user ID (UUID).
- **Responses:**
 - 200 OK: User details.
 - 404 Not Found: User not found.

Update User

- **Endpoint:** /api/users/{userId}
- **Method:** PUT
- **Summary:** Updates an existing user.
- **Parameters:**
 - userId (path, required): The user ID (UUID).
- **Request Body:** UpdateUserRequest
- **Responses:**
 - 200 OK: User updated.
 - 400 Bad Request: Invalid request.
 - 404 Not Found: User not found.

Delete User

- **Endpoint:** /api/users/{userId}
- **Method:** DELETE
- **Summary:** Deletes a user by their ID.
- **Parameters:**
 - userId (path, required): The user ID (UUID).
- **Responses:**
 - 200 OK: User deleted.
 - 404 Not Found: User not found.

Get All Users

- **Endpoint:** /api/users
- **Method:** GET
- **Summary:** Gets all users.
- **Responses:**
 - 200 OK: List of users.

Create User

- **Endpoint:** /api/users

- **Method:** POST
- **Summary:** Creates a new user.
- **Request Body:** CreateUserRequest
- **Responses:**
 - 201 Created: User created.
 - 400 Bad Request: Invalid request.

Account Management

Get Account by ID

- **Endpoint:** /api/accounts/{id}
- **Method:** GET
- **Summary:** Gets an account by its ID.
- **Parameters:**
 - id (path, required): The account ID (UUID).
- **Responses:**
 - 200 OK: Account details.
 - 404 Not Found: Account not found.

Create Checking Account

- **Endpoint:** /api/accounts/checking
- **Method:** POST
- **Summary:** Creates a new checking account.
- **Request Body:** CreateCheckingAccountRequest
- **Responses:**
 - 201 Created: Checking account created.
 - 400 Bad Request: Invalid request.

Create Saving Account

- **Endpoint:** /api/accounts/saving
- **Method:** POST

- **Summary:** Creates a new saving account.
- **Request Body:** CreateSavingAccountRequest
- **Responses:**
 - 201 Created: Saving account created.
 - 400 Bad Request: Invalid request.

Account Transactions

Deposit into Account

- **Endpoint:** /api/accounts/deposit
- **Method:** POST
- **Summary:** Deposits an amount into an account.
- **Request Body:** DepositRequest
- **Responses:**
 - 200 OK: Deposit successful.
 - 400 Bad Request: Invalid request.
 - 404 Not Found: Account not found.

Withdraw from Account

- **Endpoint:** /api/accounts/withdraw
- **Method:** POST
- **Summary:** Withdraws an amount from an account.
- **Request Body:** WithdrawRequest
- **Responses:**
 - 200 OK: Withdrawal successful.
 - 400 Bad Request: Invalid request.
 - 404 Not Found: Account not found.

Transfer between Accounts

- Endpoint: /api/accounts/transfer
- Method: POST

- Summary: Transfers an amount from one account to another.
- Request Body: TransferRequest
- Responses:
- 200 OK: Transfer successful.
- 400 Bad Request: Invalid request.
- 404 Not Found: Account not found.

Get Account Balance

- Endpoint: /api/accounts/{accountId}/balance
- Method: GET
- Summary: Gets the balance of an account.
- Parameters:
- accountId (path, required): The account ID (UUID).
- Responses:
- 200 OK: Account balance.
- 404 Not Found: Account not found.

Add Interest to Account

- **Endpoint:** /api/accounts/addInterest
- **Method:** POST
- **Summary:** Adds interest to an account for a specified number of years.
- **Parameters:**
 - accountId (query): The account ID (UUID).
 - years (query): The number of years.
- **Responses:**
 - 200 OK: Interest added.
 - 400 Bad Request: Invalid request.
 - 404 Not Found: Account not found.

Interest Calculation

- **Endpoint:** /api/accounts/InterestCalculation
- **Method:** POST
- **Parameters:**
 - accountId (query): The account ID (UUID).
 - years (query): The number of years.
- **Responses:**
 - 200 OK: Interest calculated.
 - 400 Bad Request: Invalid request.
 - 404 Not Found: Account not found.

Transactions

Get All Transactions

- **Endpoint:** /api/transaction
- **Method:** GET
- **Summary:** Gets all transactions.
- **Responses:**
 - 200 OK: List of transactions.

Get Transaction by ID

- **Endpoint:** /api/transaction/{id}
- **Method:** GET
- **Summary:** Gets a transaction by its ID.
- **Parameters:**
 - id (path, required): The transaction ID (UUID).
- **Responses:**
 - 200 OK: Transaction details.
 - 404 Not Found: Transaction not found.

Schemas

CreateCheckingAccountRequest

- **Description:** Request model for creating a checking account.
- **Properties:**
 - accountNumbers (number, required): Gets or sets the account number. Must be a 10-digit number.
 - balance (number, required): Gets or sets the balance. Must be a positive value.
 - overdrafts (number, optional): Gets or sets the overdrafts. Must be a positive value.
 - currencyCode (string, required): Gets or sets the currency code. Must be one of the following: USD, EGP, SAR, EUR, GBP.
 - userId (string, required): Gets or sets the user ID (UUID).

CreateSavingAccountRequest

- **Description:** Request model for creating a savings account.
- **Properties:**
 - accountNumbers (number, required): Gets or sets the account number. Must be a 10-digit number.
 - balance (number, required): Gets or sets the balance. Must be a positive value.
 - interest (number, required): Gets or sets the interest rate. Must be a positive value.
 - currencyCode (string, required): Gets or sets the currency code. Must be one of the following: USD, EGP, SAR, EUR, GBP.
 - userId (string, required): Gets or sets the user ID (UUID).

CreateUserRequest

- **Description:** Request model for creating a user.
- **Properties:**
 - username (string, required): Gets or sets the username. Must be between 3 and 50 characters.
 - email (string, required): Gets or sets the email address. Must be a valid email address from a valid domain.

DepositRequest

- **Properties:**
 - accountId (string, required): The account ID (UUID).
 - amount (number, required): The amount to deposit. Must be a positive value.

ProblemDetails

- **Properties:**
 - type (string, optional): The type of problem.
 - title (string, optional): The title of the problem.
 - status (integer, optional): The HTTP status code.
 - detail (string, optional): The detail of the problem.
 - instance (string, optional): The instance of the problem.

TransferRequest

- **Properties:**
 - sourceAccountId (string, required): The source account ID (UUID).
 - targetAccountId (string, required): The target account ID (UUID).
 - amount (number, required): The amount to transfer. Must be a positive value.

UpdateAccountRequest

- **Properties:**
 - balance (number, required): The new balance. Must be a positive value.

UpdateUserRequest

- **Properties:**
 - username (string, optional): The new username. Must be between 3 and 50 characters.
 - email (string, optional): The new email address. Must be a valid email address from a valid domain.

WithdrawRequest

- **Properties:**
 - accountId (string, required): The account ID (UUID).
 - amount (number, required): The amount to withdraw. Must be a positive value.